

▼ Get the Data

Ecommerce Customers csv file from the company. It has Customer info, such as Email, Address, and their color Avatar. Then it also has numerical value columns:

- Avg. Session Length: Average session of in-store style advice sessions.
- Time on App: Average time spent on App in minutes
- Time on Website: Average time spent on Website in minutes
- Length of Membership: How many years the customer has been a member.

```
[5]: df=pd.read_csv('Ecommerce Customers')
```

Check the head of customers, and check out its info() and describe() methods.

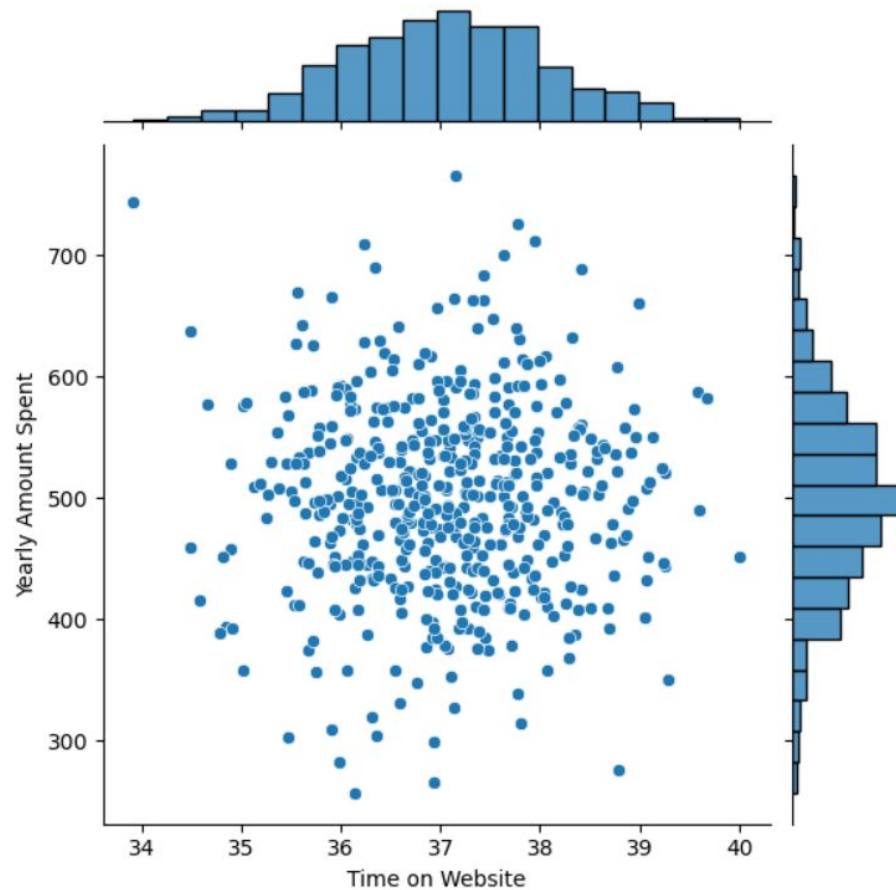
```
[6]: df.head()
```

[6]:	Email	Address	Avatar	Avg. Session Length	Time on App	Time on Website	Length of Membership	Yearly Amount Spent
0	mstephenson@fernandez.com	835 Frank Tunnel\nWrightmouth, MI 82180-9605	Violet	34.497268	12.655651	39.577668	4.082621	587.951054
1	hduke@hotmail.com	4547 Archer Common\nDiazchester, CA 06566-8576	DarkGreen	31.926272	11.109461	37.268959	2.664034	392.204933
2	pallen@yahoo.com	24645 Valerie Unions Suite 582\nCobbborough, D...	Bisque	33.000915	11.330278	37.110597	4.104543	487.547505
3	riverarebecca@gmail.com	1414 David Throughway\nPort Jason, OH 22070-1220	SaddleBrown	34.305557	13.717514	36.721283	3.120179	581.852344
4	mstephens@davidson-herman.com	14023 Rodriguez Passage\nPort Jacobville, PR 3...	MediumAquaMarine	33.330673	12.795189	37.536653	4.446308	599.406092

Exploratory Data Analysis

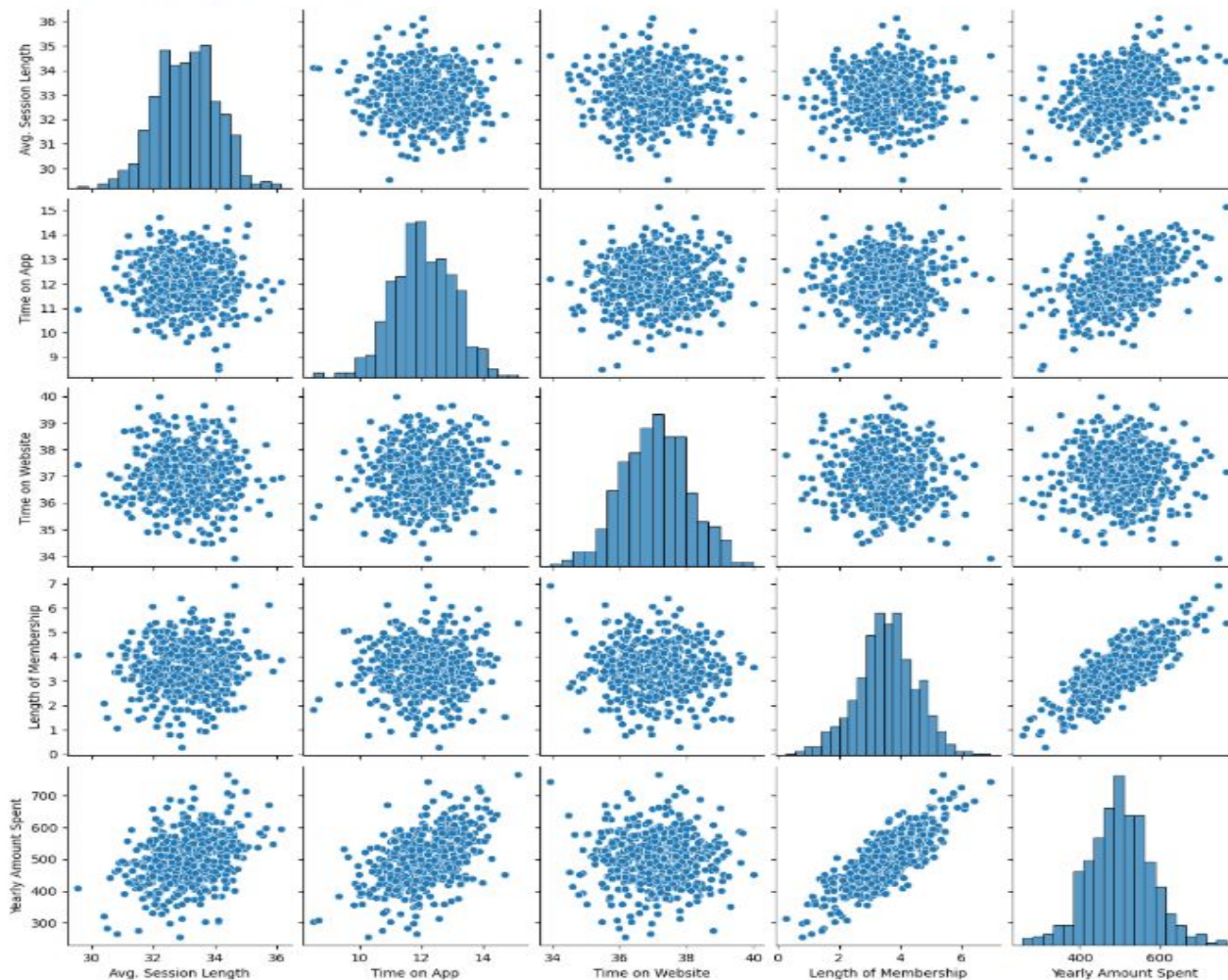
```
[9]: sns.jointplot(x='Time on Website',y='Yearly Amount Spent',data=df)
```

```
[9]: <seaborn.axisgrid.JointGrid at 0x1d287d7af90>
```



```
sns.pairplot(df)
```

```
<seaborn.axisgrid.PairGrid at 0x1d28b6c3b60>
```



▼ Training the Model

Now its time to train our model on our training data!

**** Import LinearRegression from sklearn.linear_model ****

```
[20]: from sklearn.linear_model import LinearRegression
```

Create an instance of a LinearRegression() model named lm.

```
[21]: lm=LinearRegression()
```

**** Train/fit lm on the training data.****

```
[22]: lm.fit(X_train,y_train)
```

```
[22]: ▼ LinearRegression ⓘ ⓘ  
      ► Parameters
```

Print out the coefficients of the model

```
[23]: lm.coef_
```

```
[23]: array([25.98154972, 38.59015875,  0.19040528, 61.27909654])
```

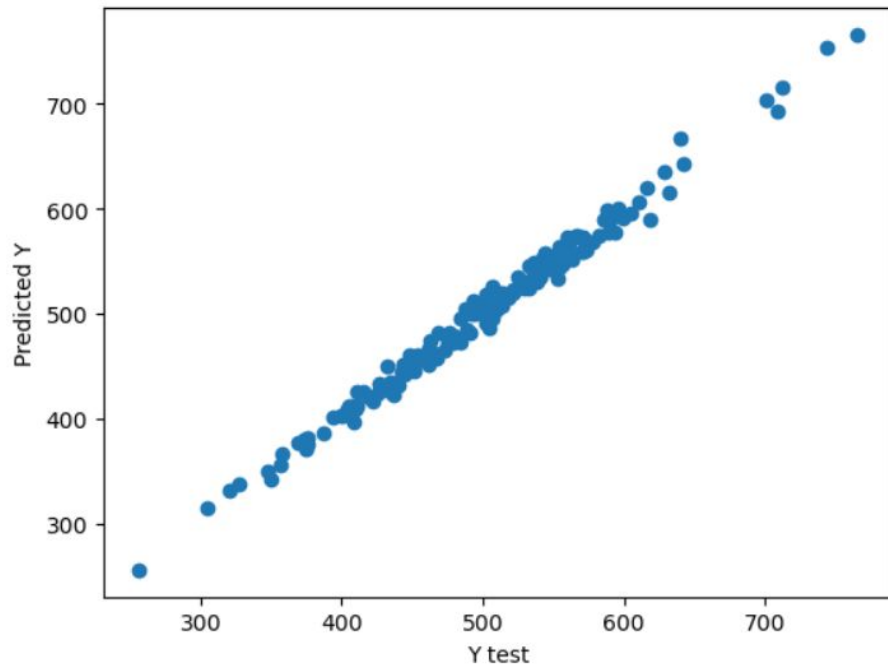
Predicting Test Data

```
24]: predictions = lm.predict(X_test)
```

**** Create a scatterplot of the real test values versus the predicted values. ****

```
25]: plt.scatter(y_test, predictions)
plt.xlabel('Y test')
plt.ylabel('Predicted Y')
```

```
25]: Text(0, 0.5, 'Predicted Y')
```



Evaluating the Model

```
[27]: from sklearn import metrics

print('MAE: ',metrics.mean_absolute_error(y_test,predictions))
print('MSE: ',metrics.mean_squared_error(y_test,predictions))
print('RMSE: ',np.sqrt(metrics.mean_squared_error(y_test,predictions)))

MAE: 7.228148653430804
MSE: 79.81305165097386
RMSE: 2.688521648309867
```

Residuals

```
[29]: sns.distplot(y_test-predictions,bins=50)
```

C:\Users\merve\AppData\Local\Temp\ipykernel_17912\1019884033.py:1: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see <https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751>

```
sns.distplot(y_test-predictions,bins=50)
```

```
[29]: <Axes: xlabel='Yearly Amount Spent', ylabel='Density'>
```

